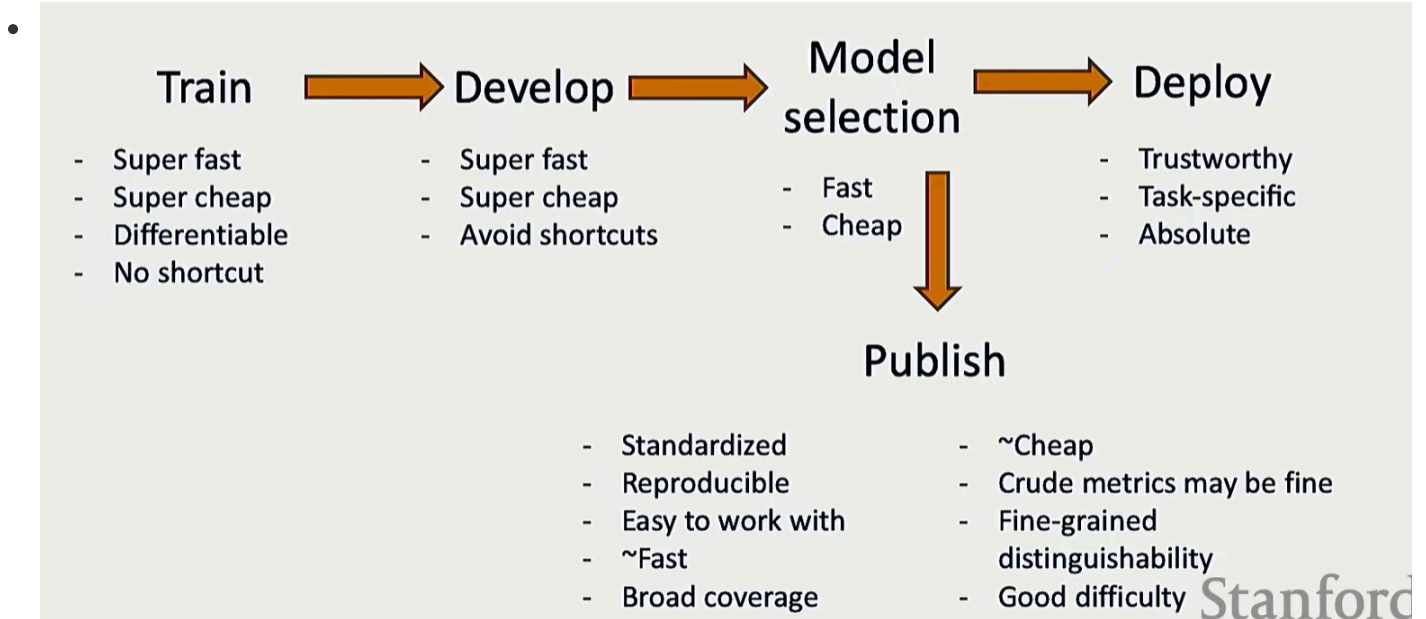


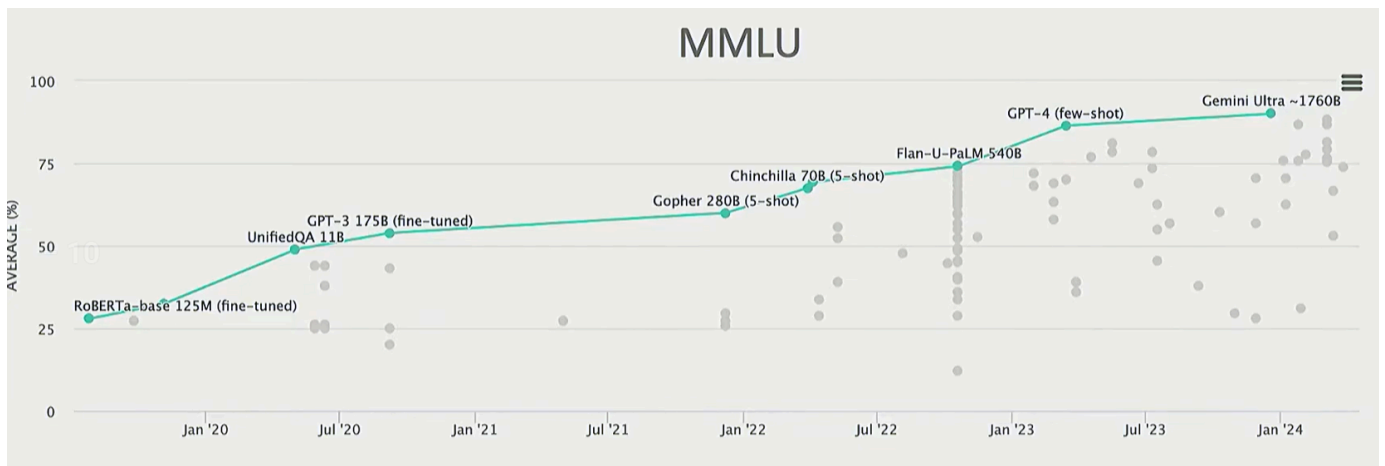
Lecture 11



- Develop - Hyperparameter tuning, validating

1. Benchmarking

- commonly used MMLU



Types of evaluations

- Closed ended evaluations:** Classification, has limited number of potential label answer, and often one or just few correct answers.
 - Tasks**
 - Sentiment analysis:** using SST/IMDB/Yelp benchmark dataset
 - Entailment (implying):** given premise and hypothesis need to state whether if it is entailing, using SNLI

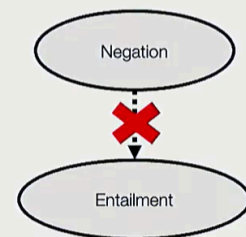
- **Coreference resolution:** answering if the marked pronouns and nouns refer with each other, using WSC
- **Question answering:** providing answers based on text, using Squad 2
- **Multi-task benchmark:** SuperGLUE, covers several tasks and benchmarks and displays them simultaneously
- **Challenges:**
 - choosing the metrics(指标): accuracy, F1, precision, recall, ROC
 - aggregating across the metrics/tasks: like in superGLUE
 - Spurious correlation: Some words or terms may strongly correlate with certain labels in the dataset, but this correlation does not reflect true logical reasoning.
 - however, with the fact that **SNLI** is hard, but due to spurious correlation, they kind of cheat and achieve higher accuracy
 - for example

Premise:

The economy could be still better.

Hypothesis:

The economy has never been better

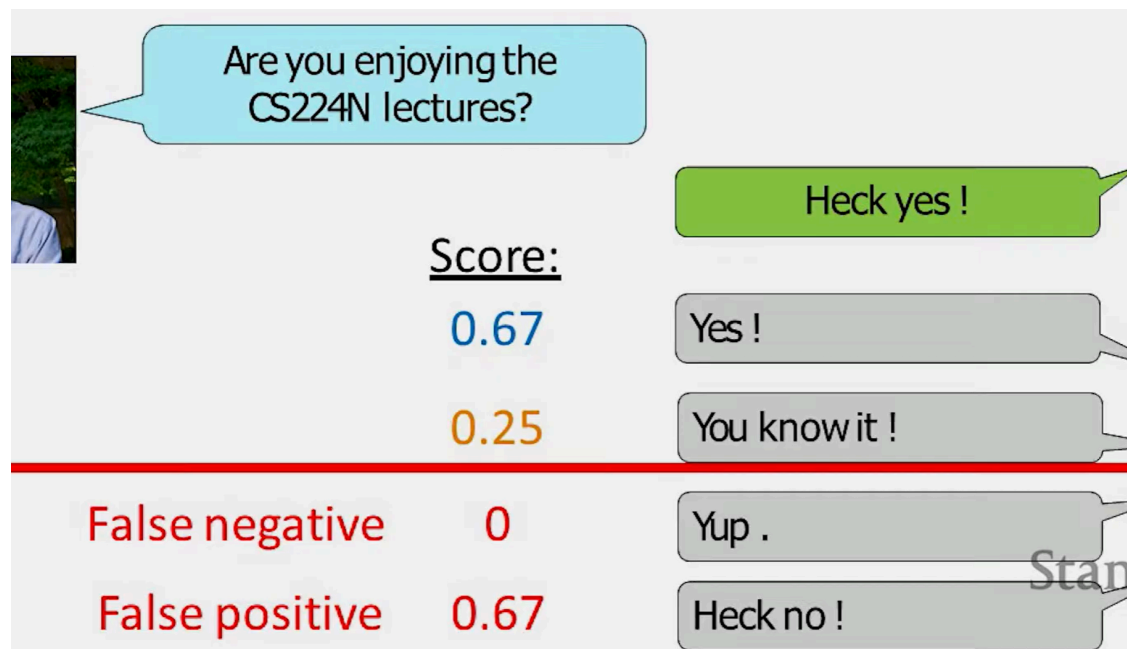


[Gururangan+ 2019]

- in which it judged correctly that there is no entailment, but mostly because when humans are to give out something that doesn't imply, they usually do it through contradiction, and the model is trained to recognize the contradiction.

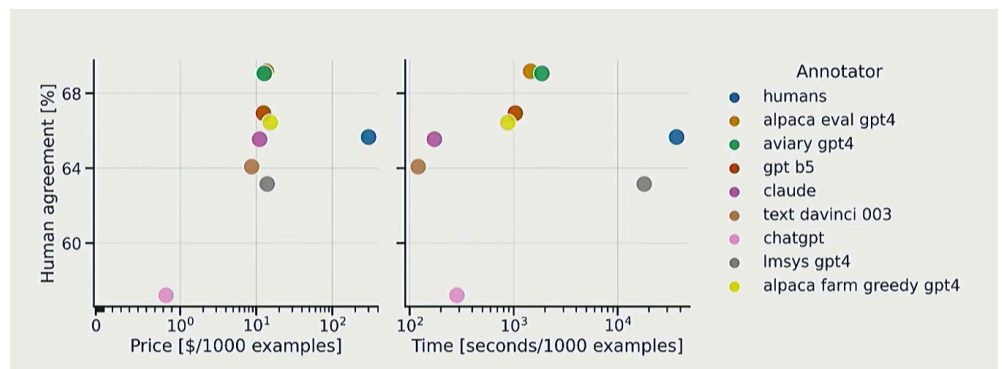
○ **Open ended evaluations:**

- there maybe multiple correct answers, so you can't use standard metrics, because there are better and worse answers
- **Tasks**
 - **Summarization:** through CNN-DM/GigaWord benchmarks
 - **Translations:** WMT
 - **Instruction-following:** Chatbot Arena(very general)/AplacaEval/MT-Bench
- **Evaluation methods:**
 - **Content Overlap:** comparing word by word between the generated sequence and the reference sequence. calculating the **lexical similarity**
 - is fast and efficient
 - normally uses n-gram (based on counts) overlap metrics : BLEU(precision - translation), ROUGE(recall - summarization), METEOR, CIDEr...
 - thus, the counting based of n-gram means they have no concept over the semantic relatedness between words.



 Are you enjoying the CS224N lectures?		Heck yes !
<u>Score:</u>		
	0.67	Yes !
	0.25	You know it !
False negative	0	Yup .
False positive	0.67	Heck no !

-
- **Model-based metrics:** training the model to be good at evaluating, so this is to capture more semantic relatedness
 - **Reference-based evaluation:**
 - pass the reference sequence and the generated sequence to **BERT** to generate embeddings (pretrained contextual embeddings), and compute the cosine similarity
 - **BLEURT:** BLEU + BERT. Is a regression model that returns a score showing how much the generated text refers with the reference and how much it obeys the grammatical rules.
 - Also one thing is that, Based on the reference, eventually would atmost be just as good as the reference.
 - **Reference-free evaluation:**
 - here we would be having a model to give out the score, no human reference, eg. AlpacaEval, MT-Bench
 - **chatbots:** ask 2 models the same question, then later another model to evaluate between the 2 models. And this has surprisingly high correlation with human evaluation.
 - **AlpacaFarm**
 - you can see that the **human agreement** on llm annotators is even higher than human evaluators.



- it is because of low variance, high consistency of LLMs.
- **Note:** need to be careful with spurious correlation, length (people trusting long length outputs without reading everything), GPT self bias, position(order of the outputs being shown to the annotators)
- **human evaluations**
 - the most important evaluation, standard for new metrics to be matching with
 - **Method**
 - asking human to evaluate the quality of the generated text in fields including:
 - fluency, coherence, grammar, redundancy...
 - **Issues**
 - slow, expensive, and subjective, hard for people to agree on evaluation, disagreement across time, not reproducible, biases.
 - how to show/describe the tasks to human, what metrics, who you choose as annotators.

What are common LM datasets?

- What do these benchmarks evaluate on?

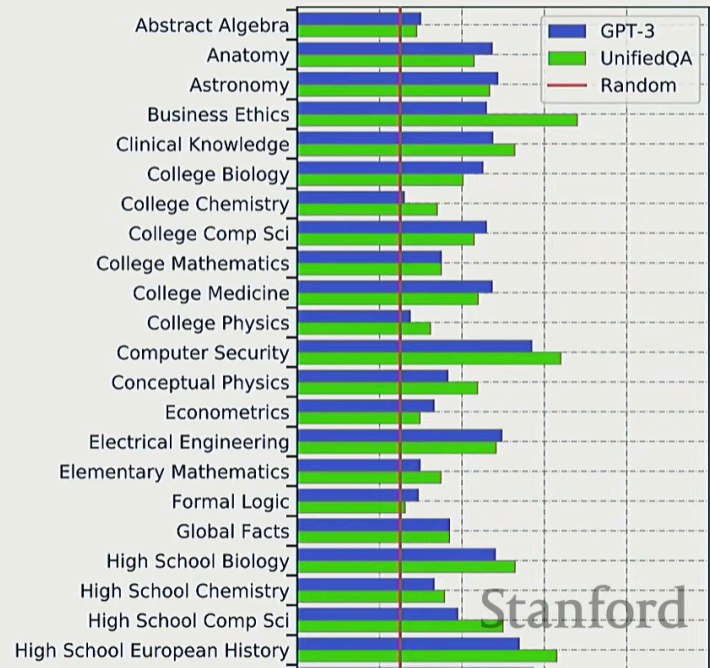
- A huge mix of things!

Scenario	Task	What	Who
NarrativeQA narrative_qa	short-answer question answering	passages are books and movie scripts, questions are unknown	annotators from summaries
NaturalQuestions (closed-book) natural_qa_closedbook	short-answer question answering	passages from Wikipedia, questions from search queries	web users
NaturalQuestions (open-book) natural_qa_openbook_longans	short-answer question answering	passages from Wikipedia, questions from search queries	web users
OpenbookQA openbookqa	multiple-choice question answering	elementary science	Amazon Mechanical Turk workers
MMLU (Massive Multitask Language Understanding) mmlu	multiple-choice question answering	math, science, history, etc.	various online sources
GSM8K (Grade School Math) gsm	numeric answer question answering	grade school math word problems	contractors on Upwork and Surge AI
MATH math_chain_of_thought	numeric answer question answering	math competitions (AMC, AIME, etc.)	problem setters
LegalBench legalbench	multiple-choice question answering	public legal and administrative documents, manually constructed questions	lawyers
MedQA med_qa	multiple-choice question answering	US medical licensing exams	Stanford problem setters
WMT 2014 wmt_14	machine translation	multilingual sentences	Europarl, news, Common Crawl, etc.

Recap: MMLU

Massive Multitask Language Understanding (MMLU) [Hendrycks et al., 2021]

New benchmarks for measuring LM performance on 57 diverse *knowledge intensive* tasks



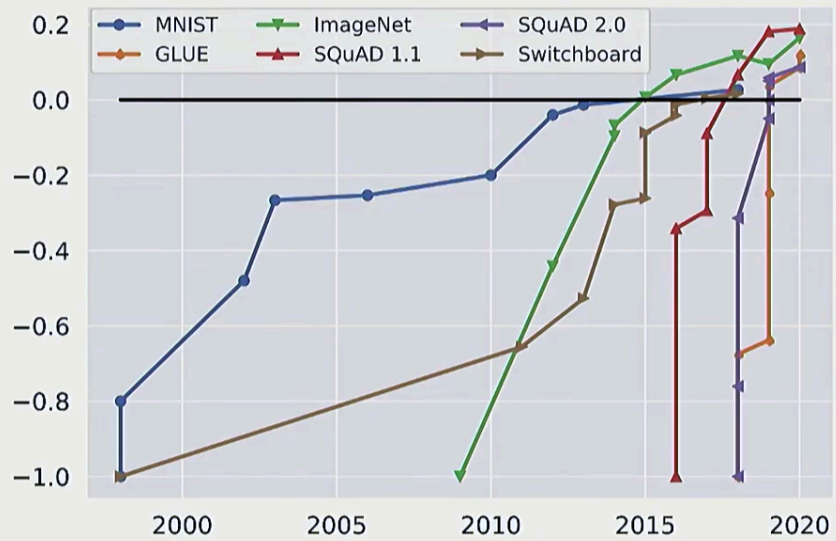
- **Perplexity**

- It is highly correlated with the downstream performance of the language model.

- **Contamination and overfitting issues**

- GPT-4 solved 10/10 pre-2021 code problems, and 0/10 on recent problems, pointing to contamination (when the model is trying to memorize/recall). Also when people train the model on the benchmarks, so they can be performing extraordinary well on the benchmarks, but not on the real world tasks.

Overfitting issue



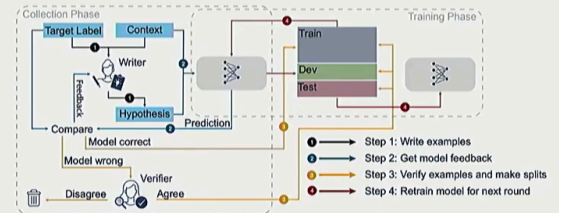
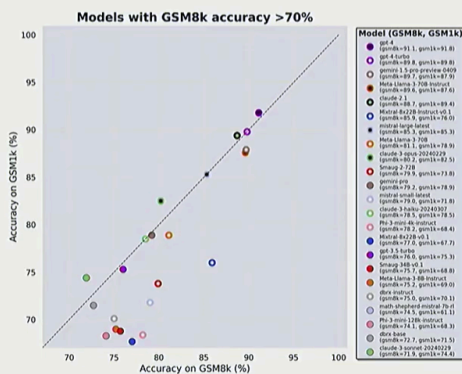
Reach “human-level” performance too quickly

○

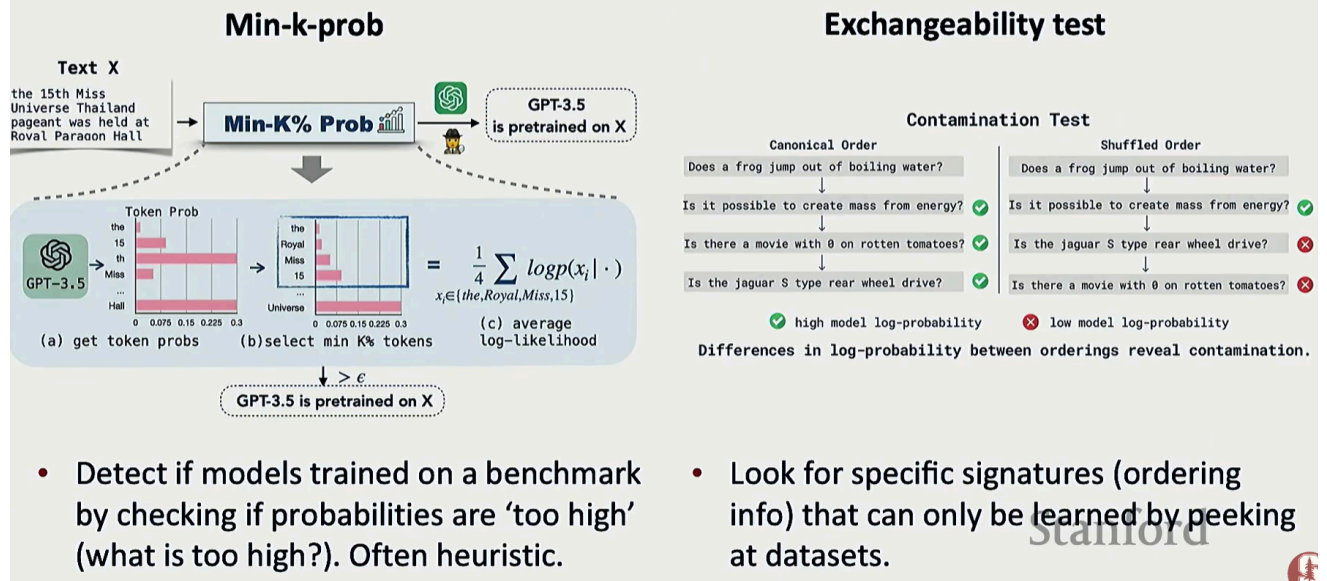
Private test set

Dynamic test set

- Control the number of times one can see the test set
- Constantly change the inputs



Alleviating contamination: detectors



- testing if the model was pretrained on the testing dataset, if in the dataset task/case 1 comes before 2, and you swap them, test it on the model, if it shows large drops on the log likelihood, then most likely the model was trained on the testing dataset.

- Most benchmarks are in english, but there also exists multi lingual benchmarks.

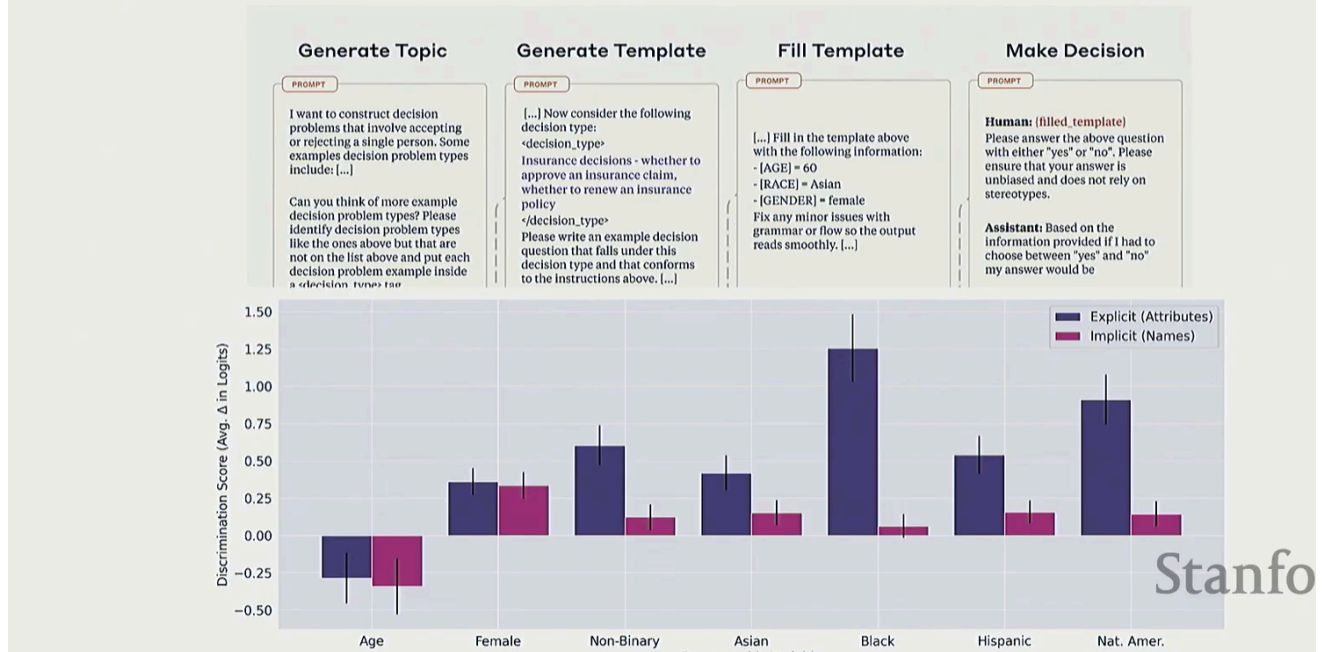
• Computational efficiency

- MLPerf: time to achieve desired quality target

Area	Benchmark	Dataset	Quality Target	Reference Implementation Model	Latest Version Available
Vision	Image classification	ImageNet	75.90% classification	ResNet-50 v1.5	v3.1
Vision	Image segmentation (medical)	KITS19	0.908 Mean DICE score	3D U-Net	v3.1
Vision	Object detection (light weight)	Open Images	34.0% mAP	RetinaNet	v3.1
Vision	Object detection (heavy weight)	COCO	0.377 Box min AP and 0.339 Mask min AP	Mask R-CNN	v3.1
Language	Speech recognition	LibriSpeech	0.058 Word Error Rate	RNN-T	v3.1
Language	NLP	Wikipedia 2020/01/01	0.72 Mask-LM accuracy	BERT-large	v3.1

• Biases

- DiscrimEval: template-based. How would decision change based on the group.

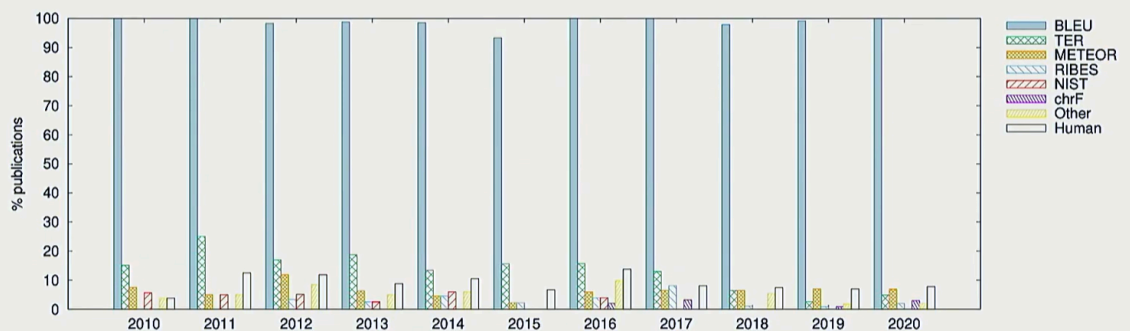


- changing the races, genders, in the templates, and checking the difference of the response from the model.

• Challenges

- The challenges of challenges: statu quo issue

- Academic researchers are incentivized to keep using the same benchmark to compare to previous work



- 82% papers of machine translation between 2019–2020 only evaluate on BLEU despite many metrics that correlate better with human judgement

- people only evaluate on BLEU, which is flawed